LED Product Testing Analysis Report

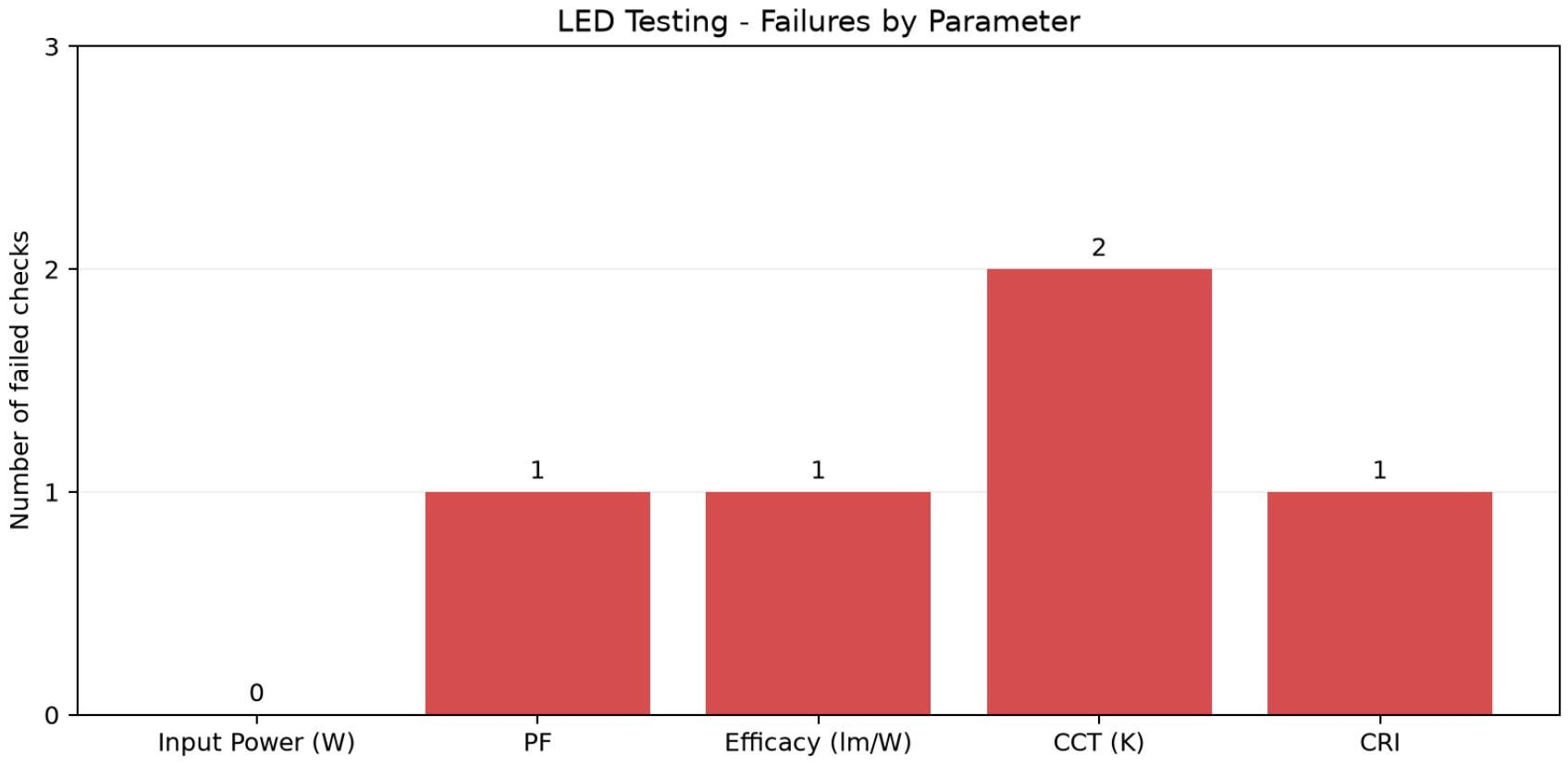
Input file: led_samples.csv
Checks: 25 | PASS: 20 | FAIL: 5 | AI status: SUCCESS

Detailed Test Results

Sample	Parameter	Measured	Specification	Status	Issue
S1	Input Power (W)	9	8.1 - 9.9	PASS	Within specification
S1	PF	0.95	0.9 - 1	PASS	Within specification
S1	Efficacy (lm/W)	105	100 - 999	PASS	Within specification
S1	CCT (K)	6500	6000 - 7000	PASS	Within specification
S1	CRI	82	80 - 100	PASS	Within specification
S2	Input Power (W)	9.4	8.1 - 9.9	PASS	Within specification
S2	PF	0.87	0.9 - 1	FAIL	Below minimum by 0.03
S2	Efficacy (lm/W)	98	100 - 999	FAIL	Below minimum by 2
S2	CCT (K)	6400	6000 - 7000	PASS	Within specification
S2	CRI	81	80 - 100	PASS	Within specification
S3	Input Power (W)	8.8	8.1 - 9.9	PASS	Within specification
S3	PF	0.92	0.9 - 1	PASS	Within specification
S3	Efficacy (lm/W)	110	100 - 999	PASS	Within specification
S3	CCT (K)	5800	6000 - 7000	FAIL	Below minimum by 200
S3	CRI	79	80 - 100	FAIL	Below minimum by 1
S4	Input Power (W)	8.5	8.1 - 9.9	PASS	Within specification
S4	PF	0.93	0.9 - 1	PASS	Within specification
S4	Efficacy (lm/W)	102	100 - 999	PASS	Within specification

Sample	Parameter	Measured	Specification	Status	Issue
S4	CCT (K)	5600	6000 - 7000	FAIL	Below minimum by 400
S4	CRI	81	80 - 100	PASS	Within specification
S5	Input Power (W)	8.6	8.1 - 9.9	PASS	Within specification
S5	PF	0.91	0.9 - 1	PASS	Within specification
S5	Efficacy (lm/W)	105	100 - 999	PASS	Within specification
S5	CCT (K)	6200	6000 - 7000	PASS	Within specification
S5	CRI	83	80 - 100	PASS	Within specification

Failure Analysis Graph



AI-Assisted Technical Analysis

Note: The root causes, verification steps, and corrective actions detailed below are preliminary engineering hypotheses based on the provided test data. A qualified product development engineer must physically verify these recommendations in the laboratory before implementing design or manufacturing changes.

Failure Analysis & Recommendations

1. Sample S2

* **Failures Explained:**

* **Power Factor (PF):** Measured **0.87** (Spec: **0.90 – 1.00**; failed by -0.03).

* **Efficacy:** Measured **98.00 lm/W** (Spec: **100 – 999 lm/W**; failed by -2.00 lm/W).

* **Plausible Technical Causes (Hypotheses):**

* *PF Issue:* Inadequate passive/active Power Factor Correction (PFC) stage tuning in the LED driver, or improper filter capacitor sizing causing Phase shift/harmonic distortion.

* *Efficacy Issue:* Electrical losses in the low-PF driver converting energy to heat rather than light, poor thermal management raising junction temperature (T_j), or lower-grade LED flux binning.

* **Practical Verification Steps:**

1. Measure driver-only efficiency and input current harmonics (THD) using a power analyzer.

2. Perform thermal imaging (thermography) on the LED board and driver to measure steady-state temperature (T_s / T_j).

* **Possible Corrective Actions:**

* Re-tune driver PFC circuit or select a driver with guaranteed higher PF (>0.95).

* Upgrade thermal interface material (TIM) or heatsink to reduce T_j.

* Move to a higher luminous flux LED bin.

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2. Sample S3

* **Failures Explained:**

* **CCT:** Measured **5800.00 K** (Spec: **6000 – 7000 K**; failed by -200 K).

* **CRI:** Measured **79.00** (Spec: **80 – 100**; failed by -1).

* **Plausible Technical Causes (Hypotheses):**

* *CCT & CRI Issue:* Over-dispersion or incorrect mix of yellow/red phosphor over the blue LED die, causing a warmer-than-intended color shift and slight degradation in color rendering completeness (R_a).

* *Binning Issue:* Sub-tier or mislabeled LED reel lot delivered by supplier.

* **Practical Verification Steps:**

1. Test bare LEDs on an integrating sphere without secondary optics or diffusers to isolate component-level performance.

2. Inspect supplier Lot/Reel material certificates (CoC) and spectral power distribution (SPD) plots for R_1–R_8 breakdown.

* **Possible Corrective Actions:**

* Tighten incoming inspection and specify higher nominal CCT/CRI bins (e.g., target 6500 K / CRI 82 minimum).

* Reject/return out-of-spec component lots to the vendor.

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3. Sample S4

* **Failures Explained:**

* **CCT:** Measured **5600.00 K** (Spec: **6000 – 7000 K**; failed by -400 K).

* **Plausible Technical Causes (Hypotheses):**

* *Phosphor Layering:* Excessive phosphor coating thickness causing strong blue-light conversion to yellow/green wavelength, shifting CCT significantly down.

* *Thermal Shift:* Excessive operational thermal drive causing red-shifting of the emission spectrum under thermal equilibrium.

* **Practical Verification Steps:**

1. Measure instantaneous ("cold pulse" @ 25°C) CCT vs. steady-state ("hot") CCT to measure thermal CCT drift.

2. Audit SMT line setup to ensure correct LED reel part numbers were mounted.

* **Possible Corrective Actions:**

* Adjust LED Purchasing Specification to shift the nominal CCT bin higher (e.g., 6500 K center target).

* Improve LED thermal path if CCT droop is confirmed to be thermal-driven.

Note: Specification decisions are computed locally. AI suggestions are not verified diagnoses and require engineering review.